Ayan Chattopadhyay

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OBJECTIVE

A computational quantum theorist by training, proficient in developing mathematical models to study complex systems & implementing robust codes for their analyses, using advanced machine learning & optimization techniques. Looking to utilize these mathematical & programming skills to solve challenging problems in quantitative finance.

EDUCATION

Princeton University, *Princeton, NJ*
PhD in Theoretical & Quantum Chemistry *Advisor: Prof. Herschel Rabitz* *Sep. 2014 – May. 2020*
RESEARCH AREAS: Machine Learning & Optimization: Algorithm development, Quantum Physics, Nonlinear Optics, Applied Mathematics: Linear Algebra
GPA (Major): 3.70 / 4.00 (Overall): 3.52 / 4.00

Indian Institute of Technology, Kharagpur, *Kharagpur, India*
Masters & Bachelors (Honors) in Chemistry *Advisor: Prof. P. K. Chattaraj* *Aug. 2009 – Apr. 2014*
GPA (Major): 8.89 / 10.00 (Overall): 8.72 / 10.00

TECHNICAL SKILLS

Mathematics: Probability Theory, Linear Algebra, Optimization (Convex & Nonlinear), Numerical Techniques.
Machine Learning: Linear Models, Dimensionality Reduction, Neural Networks, Genetic Algorithms, Gradient Algorithms.
Programming: Python, C, C++, R, Mathematica, Parallel Computing in OpenMP.
Data Science Tools: Tensorflow, Keras, Scikit-Learn, Pandas.

RELEVANT COURSES

Linear System Theory; Abstract Algebra; Mathematical Methods for Engineering Analysis; Optimal Control & Estimation; Quantum Mechanics & Quantum Computation (edX Certificate); Numerical Solutions of ODEs & PDEs.

RESEARCH PROJECTS & PUBLICATIONS

Theory of nonlinear optical spectroscopy with Optical Frequency Combs and its applications *PRINCETON UNIV., NJ*
- Derived complete theoretical model utilizing complex analysis & quantum perturbation theory.
- Performed complete theoretical analysis utilizing linear algebra & matrix decomposition tools.
- Implemented the theory in Python, C, OpenMP & NLopt (nonlinear optimization library in C). Verified complex Cauchy integrals in Mathematica. Implemented a "quantum" Gradient Descent (in C) and a Genetic Algorithm in DEAP (Python).
- *Publication*: "Efficient non-linear optical spectroscopy with multiple high-frequency modulated optical frequency combs" *Optical Society of America, Frontiers in Optics*. (Finalist: Emil Wolf Outstanding Paper Competition)
- (*Manuscript available*) "Theory for optical-frequency-comb-based nonlinear spectroscopy"
- (*In preparation*) "Discrimination of spectrally similar sensors with frequency-comb-based nonlinear spectroscopy"

Raman spectroscopy assisted discrimination of quantum systems *PRINCETON UNIV., NJ*
- Derived theoretical model utilizing quantum dynamics and control theory to guide related laser experiments.
- Implemented the theory in Python, C & OpenMP, fit related experimental data with optimization routines in C.
- (*Manuscript available*) "Reduction of spectral cross-talk in optogenetics with Raman assisted optical control"

Master's Thesis: Zero quantum potential limit of a driven double-well oscillator *I.I.T. KHARAGPUR, INDIA*
- Derived theoretical model using Bohmian quantum dynamics & implemented it using numerical techniques in FORTRAN.

Coupled Cluster Theory of Raman Optical Activity (***Summer Internship '13***) *VIRGINIA TECH, VA*
- Derived complete mathematical model using quantum perturbation theory & implemented it in C++.

Path Integral Monte Carlo & Vibronic Coupling Theory modeling of non-adiabatic systems. *UNIV. WATERLOO, ON*
- Implemented a quantum variant of Monte Carlo algorithm in FORTRAN to study special quantum systems.

AWARDS

Program in Plasma Science & Technology, *Princeton, NJ*
Research Fellowship: Summers 2018, 2019; Academic Year 2018-19 *May. 2018 – Aug. 2019*

Robert S. Hilbert Memorial Award, Optical Society of America, *Washington D.C.*
OSA Student Award, Frontiers in Optics + Laser Science, 2018 Meeting *Sep. 2018*

Jean Bennett Memorial Award, Optical Society of America, *Washington D.C.*
OSA Student Award, Frontiers in Optics + Laser Science, 2018 Meeting, [declined] *Sep. 2018*

Taylor Fellowship, *Princeton, NJ*
Academic Fellowship, Dept. of Chemistry, Princeton University *Sep. 2017 – May. 2018*

INSPIRE Scholarship, *IIT Kharagpur, WB, India*
Academic Fellowship, Dept. of Science & Technology, India *Aug. 2009 – May. 2014*